

RESEARCH ARTICLE

Combined exposure to lead and microplastics increased risk of glucose metabolism in mice via the Nrf2/NF- κ B pathway

Mengqiang Zhu¹ | Peng Li¹ | Tongfen Xu² | Guoyun Zhang¹ | Zhuo Xu² |
Xiangrong Wang¹ | Lulu Zhao¹ | Haibo Yang² 

¹Linyi Hedong District Maternal and Child Health Care Hospital, Linyi, China

²Linyi People's Hospital, Linyi, China

Correspondence

Lulu Zhao, Linyi Hedong District Maternal and Child Health Care Hospital, Linyi, 276034, China.

Email: daxuesheng051@163.com

Haibo Yang, Linyi People's Hospital, Linyi, 276000, China.

Email: yanghaibocmu@163.com

Funding information

Shandong Provincial Medicine and Health Technology Development Program, Grant/Award Number: 202103061021; Key Research and Development Program of Linyi City (Medical) Project, Grant/Award Numbers: 2022YX0112, 2022YX0109

[Correction added on 13 February 2024 after first online publication: Affiliations were updated.]

Abstract

The purpose of this study was to explore the effects of combined lead (Pb) and two types of microplastic (MP) (polyvinyl chloride [PVC] and polyethylene [PE]) exposure on glucose metabolism and investigate the role of the nuclear factor erythroid 2-related factor 2 (Nrf2)/nuclear factor-kappa B (NF- κ B) signaling pathway in mediating these effects in mice. Adult C57BL/6J mice were randomly divided into four groups: control, Pb (100 mg/L), MPs (containing 10 mg/L PE and PVC), and Pb + MPs, each of which was treated with drinking water. Treatments were conducted for 6 weeks. Co-exposure to Pb + MPs exhibited increase glycosylated serum protein levels, insulin resistance, and damaged glucose tolerance compared with the control mice. Additionally, treatment with Pb + MPs caused more severe damage to hepatocytes than when exposed to them alone concomitantly, exposed to Pb + MPs exhibited improved the levels of interleukin-6, tumor necrosis factor-alpha, and malondialdehyde, but reduced superoxide dismutase, glutathione peroxidase, and catalase assay in livers. Furthermore, they increase the Kelch-like ECH-associated protein 1 (Keap1) and phosphorylated p-NF- κ B protein levels but reduced the protein levels of heme oxygenase-1 and Nrf2, as well as increased Keap1 mRNA and Nrf2 mRNA. Co-exposure to Pb + MP impacts glucose metabolism via the Nrf2 /NF- κ B pathway.

KEYWORDS

glucose metabolism, lead, liver, microplastics, oxidative stress

1 | INTRODUCTION

Lead (Pb) and microplastics (MPs) are two widespread environmental contaminants with potential adverse effects on the environment and human health. Pb is found in various sources such as insecticides, glass, paints, ceramics, water pipes, and historically in leaded gasoline.¹ By 2030, it is estimated that the global generation of electronic waste primarily originating from discarded electronic devices will reach a staggering 74 million metric tons.² Pb is an important component of electronic waste that can easily leach into groundwater and the ecosystems. It can also contaminate dust, soil, and food matrices, thereby impacting the environment and human health.³

Studies have indicated that elevated levels of Pb in blood and urine continue to be a concern among the general population today.⁴ People can consume excessive amounts of Pb by inhaling contaminated air, consuming contaminated food and water, and smoking.⁵ Autopsy studies conducted on individuals exposed Pb have revealed that the liver tissue serves as the primary storage site for Pb accounting for approximately 33% of the total burden.⁶ Although Pb is an endocrine disrupting chemical, few studies have investigated the impact of low-dose Pb exposure on the levels of blood glucose. Recently, emerging evidence indicated that Pb exposure may induce the increased production of hepatic glucose and eventually develop into type 2 diabetes mellitus (T2DM).⁷

The prevalence of diabetes in adults aged 20–79 years has been estimated at 537 million people in 2021, and this number is projected to reach 783 million by 2045. It is predicted that more than 600 million people will suffer from diabetes (11.3% of the total population) by 2030, and by 2045, the number of patients will increase to 783 million (12.2%).⁸ Despite the fact that numerous studies had addressed the relationship between T2DM and individual metal exposures, research on the impact of exposure to mixtures of metals and other environmental pollutants on health is very limited, as well as the molecular mechanisms underlying their impact on T2DM pathogenesis.

MPs, small plastic particles (≤ 5 mm in diameter), have become ubiquitous in the environmental problems globally, posing threats to ecosystems and human health. Approximately 500 000 tons of MPs were emitted worldwide in 1960.⁹ By 2017, emissions had grown exponentially, reaching 348 million tons, and half of the plastic produced annually was disposable plastic.¹⁰ Plastic particles have been found in various environments, including soil, freshwater, seawater, air, and within organisms.^{11,12} Environmental MPs can enter the body through the esophagus, respiratory, or dermal, and oral exposure is the main pathway, such as drinking water, seafood, salts, milk, and meat.¹³ Human feces have been found to contain MPs with concentrations up to 36 particles g^{-1} .¹²

Although it has been confirmed that the human body contains MPs, further toxicological research is needed to understand the potential health effects caused by MPs. Plastics are primarily classified into two types: thermosets and thermoplastics, among them, the major commodity plastics commonly found in MPs are thermoplastics, including polystyrene (PS), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polyethylene (PE), and polypropylene (PP).¹⁴ These five types of plastics are also accounted for more than 70% of European plastics demand.¹⁵ Research indicates that 5 or 20 μM PS exposure can accumulate and induce oxidative stress in the liver.¹⁶ However, the research available on the specific effects of PE and PVC is very limited.

PVC as one of the most common MPs, is widely used in many sectors, for example, in the main PVC market, including packaging, construction, and electrical applications; flexible PVC is also used in many industries, such as food and beverage.¹⁷ PE, as a common and typical MP, is one of the most extensively used polymers with the basic repeating unit. In 2018, global production of PE exceeded 100 million tons, and the PE market will grow to a size of US \$143 billion by 2026.¹⁸ Although MP pollution has gained global attention, the present study on MPs mainly focuses on their migration and distribution in the environment. However, few researches have investigated the toxicity of PVC and PE *in vivo*.

More importantly, in real-life scenarios, humans are often exposed to multiple types of MPs and heavy metals simultaneously. However, their combined effects and the underlying molecular mechanisms remain poorly understood. The nuclear factor erythroid-2-related factor-2 (Nrf2), as a transcription factor responsible for maintaining redox homeostasis, can inhibit nuclear factor kappa-B (NF- κ B), thereby exerting indirect antioxidant and antiinflammatory effects.^{19,20} Research has shown that both Nrf2 and NF- κ B may play

significant roles in toxicological impact on hepatic glucose metabolism processes and insulin sensitivity.²¹

The Nrf2/NF- κ B pathway has emerged as a potential regulatory pathway in the context of Pb and MP exposure and its impact on glucose homeostasis. Therefore, this study focuses on the effects of combined exposure to Pb and two types of MP (PVC and PE) on glucose metabolism, and elucidates the effects of Nrf2/NF- κ B pathway in this process. By elucidating the molecular mechanisms involved, we hope to enhance our understanding of the potential risks associated with environmental exposure to Pb and MPs and shed light on novel therapeutic targets for managing glucose-related metabolic disorders. The present study could provide information to elucidate complex interplay between environmental contaminants, cellular signaling pathways, and metabolic health.

2 | MATERIALS AND METHODS

2.1 | Chemicals and reagents

Polyethylene (6.5 μm) and polyvinylchlorid (6.5 μm) were provided by Huachuang Plastic Co., Ltd. (Dongguan, China). Lead (II) acetate basic ($\text{C}_4\text{H}_8\text{O}_6\text{Pb}_2$) was provided by Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). Enzyme-linked immunosorbent assay (ELISA) kits for mice tumor necrosis factor- α (TNF- α) and IL-6 were obtained from Meimian Biotechnology (Jiangsu, China). Rabbit Anti-Nrf2 antibody, Anti-HO antibody, and Kelch-like ECH-associated protein 1 (Keap-1) antibody were purchased from Beijing Biosynthesis Biotechnology Co., Ltd (Beijing, China). Anti-NF- κ B p65 antibody and Anti-p-NF κ B p65 antibody were provided by Santa Cruz Biotechnology, Inc. Malondialdehyde (MDA), superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx) assay kit were obtained from Beyotime Institute of Biotechnology, China. ELISA kit for insulin was provided by Uscn Life Science Inc. (Wuhan, China). Water used for animals was produced by a Milli-Q water purification system. All other chemicals are analytical grade.

2.2 | Animal treatment

Eight-week-old C57BL/6J mice (License No. SCXK 220-0011) were provided by Xuzhou Medical University (Xuzhou, China). The mice were kept under specific pathogen-free (SPF) conditions, and 12 h light/darkness cycles at 20–24°C and 60–70% humidity for 1 week acclimatization prior to the experiments. Animal welfare and experiments were strictly conducted following to the Guide for the National Institute of Health (NIH) guidelines for the Care and Use of Laboratory Animals, with approval from the Laboratory Animal Ethical Committee of Xuzhou Medical University.

Twenty-four mice were randomly divided into four groups (six mice in each group) according to weight, including the control group, treated with distilled water; the Pb group, fed with 100 mg/L leaded water; MPs group, fed water with a concentration of 10 mg/L PE

mixed PVC; and the Pb + MPs group, fed water with a concentration of 100 mg/L Pb and 10 mg/L PE + PVC. All animals are free to eat food and water, the body weight and fasting plasma glucose (FPG) were tested weekly at a fixed time, and the treatment lasted for 6 weeks. The Pb and MPs were prepared by dissolving them in distilled water. After the exposure time finished, the mice will be tested glucose tolerance and insulin tolerance. Afterwards, the mice were fasted for 12 h before decapitation euthanasia. Serum was obtained by centrifuging at $3000 \times g$ at 4°C for 10 min and then placed at -80°C for further analysis.

2.3 | Intraperitoneal glucose tolerance test and insulin tolerance test

Intraperitoneal glucose tolerance test (IpGTT) was performed on the 7th week of exposure. Fasting mice for 16 h before the experiment, followed by the administration of glucose (2 g/kg) through intraperitoneal injection. Blood was collected using the tail vein blood collection method. After glucose administration, glucose was measured using a blood glucose meter (Jiangsu Yuyue, China) at baseline and 15, 30, 60, 90, and 120 min. Insulin tolerance test (IpITT) was performed 2 days after IpGTT treatment. Fasting mice for 6 h before the experiment, and insulin (1 U/Kg, Novo Nordisk, Glaxo, Denmark) was injected intraperitoneally, and blood glucose was measured in the same manner as IpGTT. Calculate the area under the curve (AUC) of IpGTT and IpITT using GraphPad Prism version 5.0.

2.4 | Preparation of the liver homogenates

The liver samples from each mouse were homogenized in 10 mL/g ice-cold phosphate buffered saline (10.14 mM Na_2HPO_4 , 2.68 mM KCl, 1.76 mM KH_2PO_4 , and 136.75 mM NaCl; pH = 7.4) followed by centrifugation at $3500 \times g$ at 4°C for 15 min. The supernatants were obtained and placed at -80°C and used for later detection of antioxidants and inflammatory marker levels.

2.5 | Detection of antioxidants and inflammatory markers

All the kits used for this experiment are mouse specific kits. Lipid Peroxidation MDA Assay Kit was used to assessed the MDA levels in the liver homogenates (#S0131S, Beyotime Institute of Biotechnology, China). Total SOD assay kit with WST-8 (#S0101S, Beyotime Institute of Biotechnology, China), CAT assay kit (#S0051), and cellular GPx assay kit with NADPH (#S0056) were used to measure the levels of SOD, CAT, and GPx in liver homogenates. Liver tissue interleukin-6 (IL-6) and TNF- α were measured by ELISA kits. The ELISA kits sensitivities were 1.8 pg/mL for IL-6 and 0.99 pg/mL for TNF- α .

2.6 | Serum biochemical analysis

Biochemical analysis of alanine aminotransferase (ALT), high-density lipoprotein cholesterol (HDL-c), low-density lipoprotein cholesterol (LDL-c), serum aspartate aminotransferase (AST), cholesterol (CHOL), and triglyceride (TG) were determined using the commercially assay kits supplied by Rayto Life and Analytical Sciences Co., Ltd. (Shenzhen, China). The levels of serum insulin and glycosylated serum protein (GSP) (Changchun Huili Biotech Co., Ltd, China) were measured using commercial assay kits by Epoch (BioTek) following the provided instructions. To determined β -cell function, and insulin resistance, the homeostatic model assessment of β -cell function (HOMA- β) and homeostatic model assessment index of insulin resistance (HOMA-IR) were calculated using equations based upon fasting insulin (mmol/L) and values of fasting glucose (mIU/L). $\text{HOMA-IR} = \text{fasting glucose} \times \text{fasting insulin} / 22.5^{22}$; $\text{HOMA-}\beta = \text{fasting insulin} \times 20 / (\text{fasting glucose} - 3.5)^{23}$.

2.7 | Quantitative real-time polymerase chain reaction

Trizol (GeneBetter, Beijing, China) was used to isolate from target tissues. The quality of RNA was measured by NanoPhotometer (Implen, Eppendorf, Germany). The reverse transcriptase HiScript II Q RT SuperMix for quantitative real-time polymerase chain reaction (qPCR) (+gDNA wiper) (Vazyme, Jiangsu, China) was used to synthesize cDNA from total RNA. The primers of Keap1, Nrf2, HO-1, and GAPDH (Sangon Biotech (Shanghai) Co., Ltd.) were shown in Table 1. The reaction was performed by ABI 7500 Real-Time PCR System (Applied Biosystems, Waltham, MA). The RT-PCR cycle program parameters were: denaturation at 95°C for 30 s, followed by 40 cycles at 95°C for 5 s, and annealing of 34 s at 60°C . The selected gene expression was normalized based on the β -actin gene, and the results were calculated by the comparative CT method ($2^{-\Delta\Delta\text{CT}}$). All the RT-PCR experiments were performed in triplicate.

2.8 | Western blotting analysis

Total protein was obtained from liver by treating with RIPA buffer (10mM Na_2HPO_4 , 150mM NaCl, 1% sodium deoxycholate, 1%

TABLE 1 Primer sequences of each gene in this study.

Name	Oligo	Primer sequence
Keap1	F	CTCCGCAGAATGTTACTATCCAG
	R	ACACTGTTCAACTGGTCCTG
Nrf2	F	TCCCATTTGTAGATGACCATGAG
	R	CCATGTCCTGCTCTATGCTG
HO-1	F	AAGCCGAGAATGCTGAGTTCA
	R	GCCGTGTAGATATGGTACAAGGA
GAPDH	F	AGGTCGGTGTGAACGGATTG
	R	TGTAGACCATGTAGTTGAGGTCA

Nonidet, P-40, 1% SDS) (Beyotime Institute of Biotechnology, China). Protein was separated by polyacrylamide gel electrophoresis and transferred into polyvinylidene difluoride (PVDF) membranes (Millipore, Tercicula, CA). Then blocked the membranes at room temperature for 1 h with 10% BSA, and incubated the membranes overnight at 4°C with the primary antibody (Keap1, Nrf2, HO-1, p-NF-κB, NF-κB, and GAPDH). After incubation, the membranes were washed for three times with Tris-buffered saline with 1% Tween 20 (TBST), the membranes were incubated for 1–2 h at room temperature with the secondary anti-rabbit HRP-conjugated antibodies. The target proteins were visualized by the ECL WB detection system (Beyotime, Shanghai, China).

2.9 | Liver and pancreas histology analysis

Liver and pancreas from mice were immediately and fixed for 48 h at room temperature in 4% paraformaldehyde solution. The tissue was embedded in paraffin and cut into 5 μm thick sections, therefore, the thick sections were used for histological analysis which were stained with hematoxylin & eosin (H&E).

2.10 | Statistical analyses

Data were displayed as mean ± SD in results. Quantitative data among experimental groups were analyzed by one-way analysis of variance (ANOVA), where LSD test was used as post hoc. *p* value < .05 indicates significant difference.

3 | RESULTS

3.1 | Changes in metabolic parameters

Body weight were measured each week since the treatment begin. It was found that there was no difference in mice body weight among the four groups (Figure 1A). The fasting blood glucose of mice is measured weekly, as shown in Figure 1B, there was no significant difference in the levels of fasting blood glucose among the four groups of mice in each week. Fasting insulin and GSP were assessed to provide information about glucose metabolism and pancreatic function. Serum insulin levels had no obvious difference among the four group (Figure 1C), whereas the GSP serum levels of mice treatment with Pb + MPs were significantly higher than that in control group mice (*p* < .05).

3.2 | The effects of Pb and MPs on glucose metabolism and pancreatic islet function

Glucose tolerance reflects the efficiency of the body in processing glucose, which can be evaluated through ipGTT and ipITT. In ipGTT, the levels of blood glucose were higher in Pb + MPs group and MPs group than in control and Pb + MPs group (both *p* < .01) (Figure 2A). Similarly, in ipITT, blood glucose levels were more elevated in Pb and Pb + MPs group than in the control group (both *p* < .01); and blood glucose levels were increased in Pb + MPs group than that in MPs group (*p* < .01) (Figure 2B). The HOMA-IR index (reflects insulin resistance) was increased by 43.28% in the Pb + MPs group compared

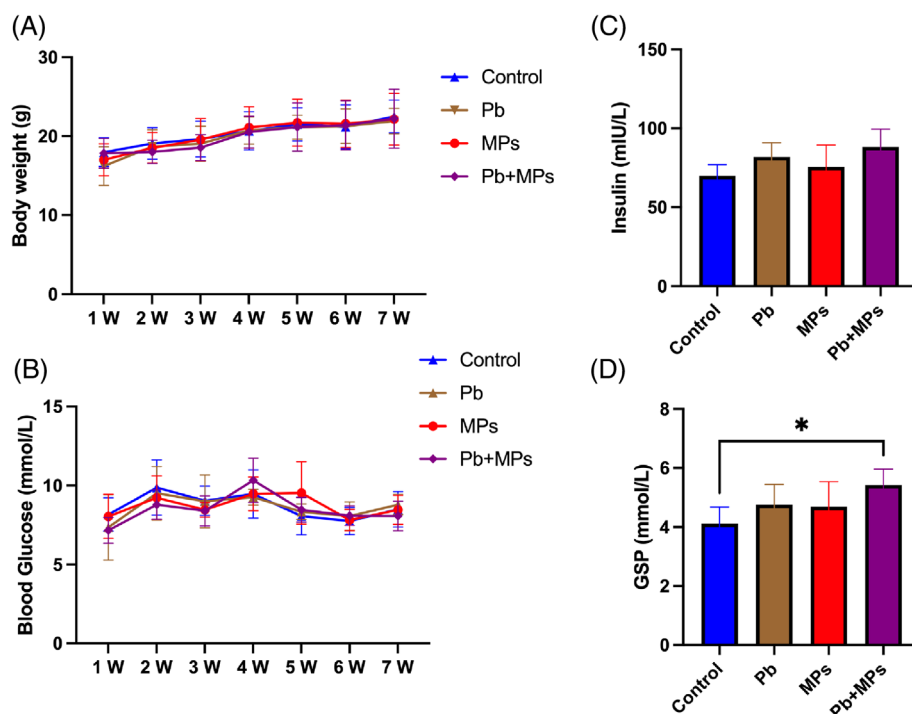


FIGURE 1 Weekly body weight, weekly blood glucose, fasting serum insulin levels, and glycosylated serum protein (GSP) of mice. (A) Weekly body weight, (B) weekly blood glucose, (C) fasting levels of serum insulin, and (D) fasting levels of serum GSP. Data are expressed as the mean ± SD, *n* = 6 in each group. There is a significant difference compared with the control group (*p* < .05).

with the control group ($p < .05$) (Figure 2C). For HOMA- β index, which is the most widely accepted theoretical index for quantifying β -cell function and mass. According to Figure 2D, there was no significant change in the level of HOMA- β index.

3.3 | Liver histological alterations exposed to Pb and MPs

The liver histopathological analysis revealed major pathological changes of mice after exposure/co-exposure to Pb and MPs (Figure 3). The liver of control mice showed normal hepatic structure,

including intact and polygonal-shape hepatocytes, large centrally located nuclei with dispersed chromatin and with slight staining acidophilic cytoplasm radiating from a central vein, and normally sized sinusoids arise in the periphery of the lobule. The Pb treatment damaged hepatic histological structure characterized by mononuclear cell infiltration, cytoplasmic loss, cytoplasmic vacuolization, and necrotic hepatocytes exhibiting pyknotic nuclei with condensed chromatin. The liver of MPs-administered mice showed mild cytoplasmic loss, immune cell infiltration, scattered necrotic hepatocytes, and congestion of hepatic tissue. The liver of Pb-combined MPs treatment mice exhibited significantly damaged hepatic histological structure, including several cytoplasmic loss and cytoplasmic vacuolization, congestion

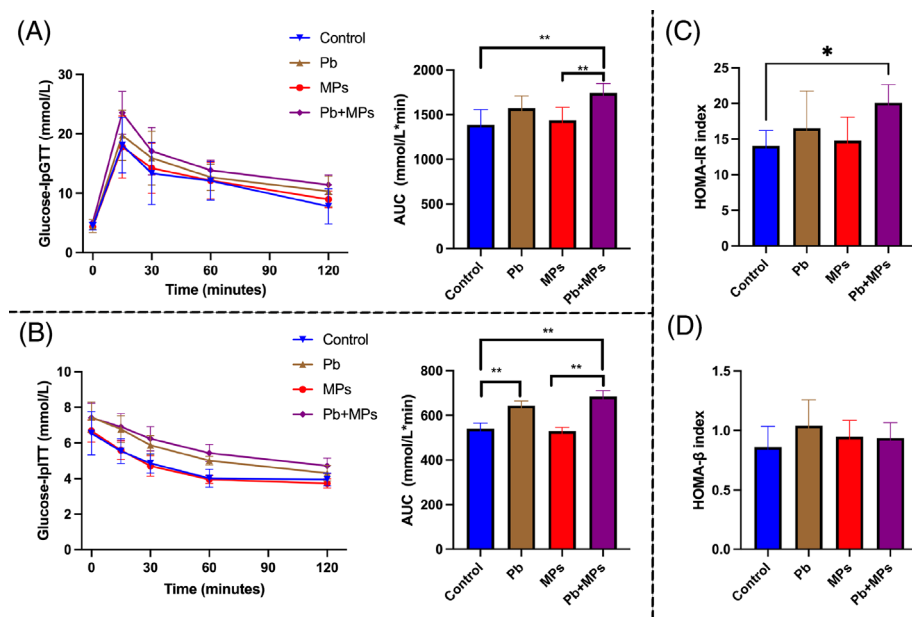


FIGURE 2 Alterations in mice intraperitoneal glucose tolerance test (IpGTT), insulin tolerance test (IpITT), homeostatic model assessment index of insulin resistance (HOMA-IR), and homeostatic model assessment of β -cell function (HOMA- β). (A) Results of IpGTT (2 g/kg BW) and the area under the curve (AUC) analysis ($n = 6$). After injection, blood glucose was tested in 0, 15, 30, 60, and 120 min. (B) IpITT (1 U/kg BW) and the AUC analysis ($n = 6$). After injection, blood glucose was tested in 0, 15, 30, 60, and 120 min. (C) HOMA-IR, HOMA-IR = fasting glucose $\times$ fasting insulin/22.5. (D) HOMA- β , HOMA- β = fasting insulin $\times$ 20/(fasting glucose-3.5).

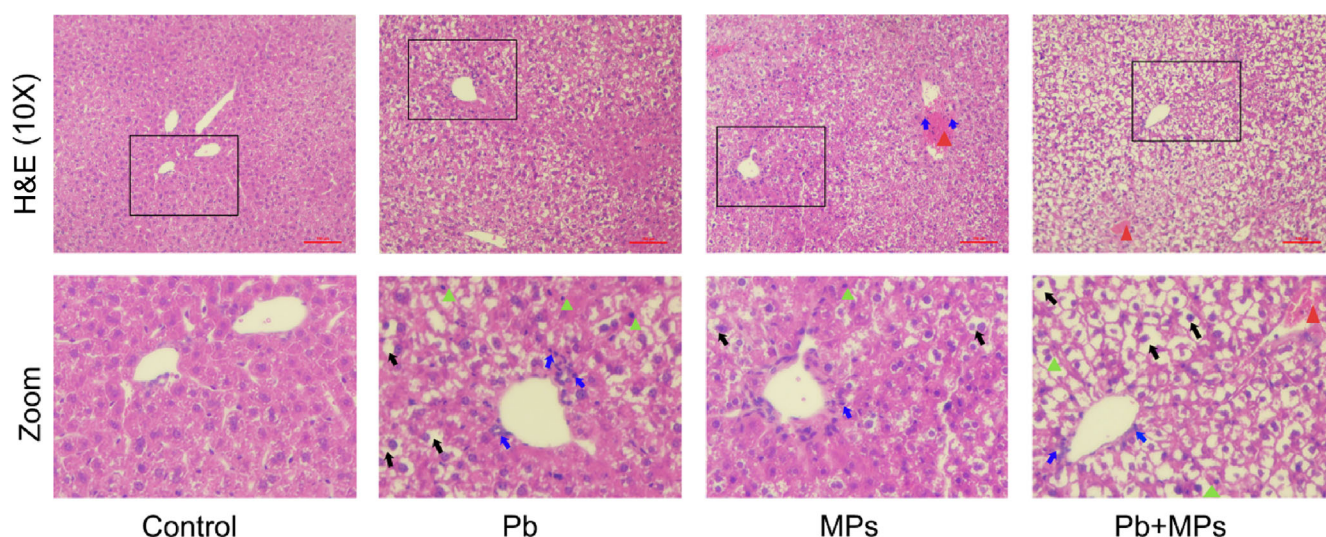


FIGURE 3 Histological images of the livers from the control, lead (Pb), microplastics (MPs), and Pb + MPs groups of mice. The image in the second line is an enlargement of the box in the first line of image. Sections were stained with hematoxylin & eosin (H&E). The magnification was $\times 100$ in all groups. Cytoplasmic loss (black arrow), immune cell infiltration (blue arrow), necrotic hepatocytes (green triangle), and congestion (red triangle).

of portal blood vessels, and accompanied by mononuclear cell infiltration and condensed chromatin.

3.4 | Pancreases histological alterations exposed to Pb and MPs

As shown in Figure 4, the mice pancreases histological features were no significant changes in the Pb and MPs exposure groups when compared with the control group. The islets of Langerhans from all groups appeared normally structure with abundant cell number. However, treated mice with Pb + MPs had vacuolations of acinar epithelial lining in pancreas (Figure 4).

3.5 | The changes in lipids serum levels of mice exposed to Pb and MPs

Next, the lipids serum levels were measured. Serum TG, CHOL, and HDL-c levels had no obvious differences among four groups, whereas the serum levels of LDL-c in Pb-MPs group were significantly increased when compared with the control group ($p < .01$) (Table 2).

3.6 | Changes in hepatic function and inflammatory markers

To evaluate the hepatic function in the Pb- and MPs-induced liver, we determined the serum ALT and AST levels of all samples. Serum ALT levels were increased after 6 weeks of exposure to Pb ($p < .05$) and Pb + MPs ($p < .01$) treatment mice as compared with the control mice, however, compared with the mice treated with Pb or MPs alone; the serum ALT levels were increased in the mice co-exposure to Pb and MPs ($p < .01$) (Figure 5A). Additionally, the AST levels were significantly increased in Pb + MP treated mice when compared with the control and MP-treated mice ($p < .01$) (Figure 5B). As shown in Figure 5C,D, exposure to Pb and mixture of Pb and MPs induced increase in the liver of TNF- α and IL-6 levels as compared with the control; meanwhile, TNF- α and IL-6 levels in mice administrated Pb and MPs alone were both lower than the mice in Pb + MPs group ($p < .01$).

3.7 | Changes in hepatic oxidative stress/antioxidant markers

To further explore the effects of Pb and MPs on the liver, alterations of oxidative stress and antioxidant markers in mice livers

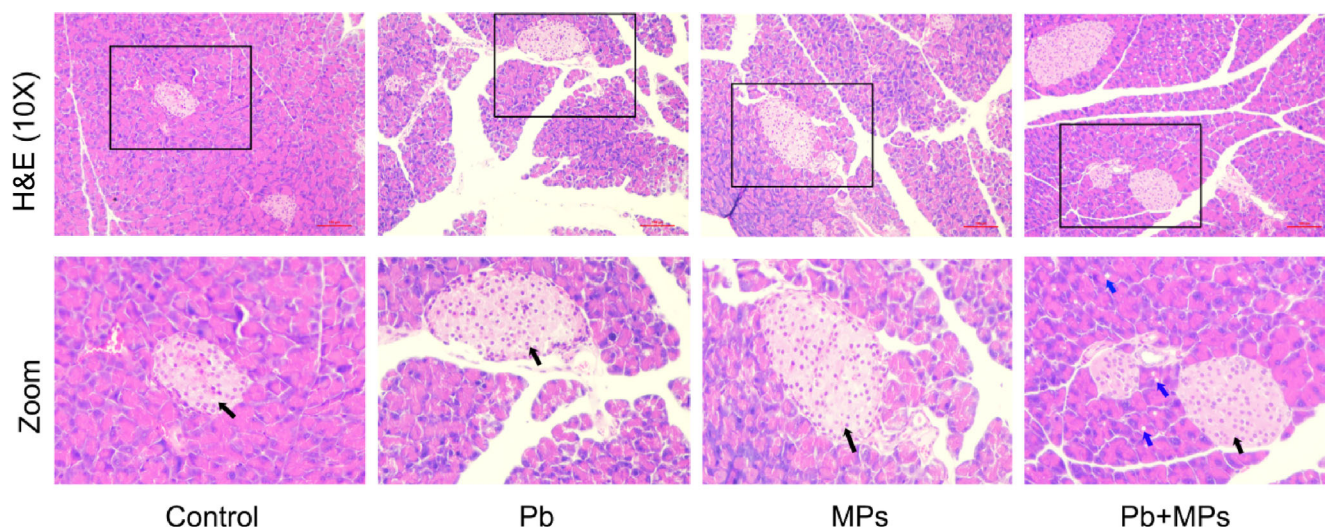


FIGURE 4 Photomicrographs of the pancreases from the control, lead (Pb), microplastics (MPs), and Pb + MPs groups of mice. Sections were stained with hematoxylin & eosin (H&E). The image in the second line is an enlargement of the box in the first line of image. Normal islets of Langerhans (black arrow) and vacuolations of pancreatic acini cells (blue arrow). The magnification was set at $\times 100$ all groups.

TABLE 2 Serum lipid profile in all groups of mice.

	Control	Pb	MPs	Pb-MPs
TG (mmol/L)	1.23 $\pm$ 0.32	1.24 $\pm$ 0.36	1.18 $\pm$ 0.2	1.46 $\pm$ 0.53
CHOL (mmol/L)	3.05 $\pm$ 0.24	3.18 $\pm$ 0.13	3.09 $\pm$ 0.19	3.24 $\pm$ 0.09
HDL-c (mmol/L)	1.92 $\pm$ 0.08	1.84 $\pm$ 0.11	1.91 $\pm$ 0.08	1.81 $\pm$ 0.13
LDL-c (mmol/L)	0.41 $\pm$ 0.06	0.47 $\pm$ 0.1	0.44 $\pm$ 0.1	0.56 $\pm$ 0.04**

Note: The results are shown as mean $\pm$ SD ($n = 6$ in each group). ** $p < .01$ significant different compared with the control group.

Abbreviations: CHOL, cholesterol; HDL-c, high-density lipoprotein cholesterol; LDL-c, low-density lipoprotein cholesterol; MPs, microplastics; Pb, lead; TG, triglyceride.

were determined, including MDA, SOD, GPx, and CAT. MDA is a genotoxic endogenous byproduct resulting from enzymatic and oxidative stress-induced lipid peroxidation. Figure 6A shows that the MDA level is significantly increased in the Pb ($p < .05$) and Pb + MPs ($p < .01$) groups when compared with the control group, and the MDA level was elevated in the Pb + MPs group when compared with the Pb and MPs alone group (both $p < .01$). On the contrary, for antioxidant markers, the SOD, GSP, and CAT levels were reduced in the Pb and Pb + MPs groups when compared with the control group, but they all increased in Pb and MPs groups compared with the Pb + MPs group ($p < .01$) (Figure 6B–D).

3.8 | Changes in hepatic Keap1/Nrf2 and NF- κ B signal pathway

As shown in Figure 7A,B, the relative protein level of Keap1 and p-NF- κ B was increased significantly in Pb and Pb + MPs than in the control group, and the relative protein level of Keap1 and p-NF- κ B in Pb + MPs group were lower as compared with Pb and MPs alone group (both $p < .01$). However, for Nrf2 and HO-1, they were decreased significantly in Pb and Pb + MPs than in the control group, but increased in Pb + MPs group when compared with Pb and MPs alone group. There was no significant difference change in the levels of NF- κ B. Figure 7C–E show the relative mRNA levels of Keap1, Nrf2,

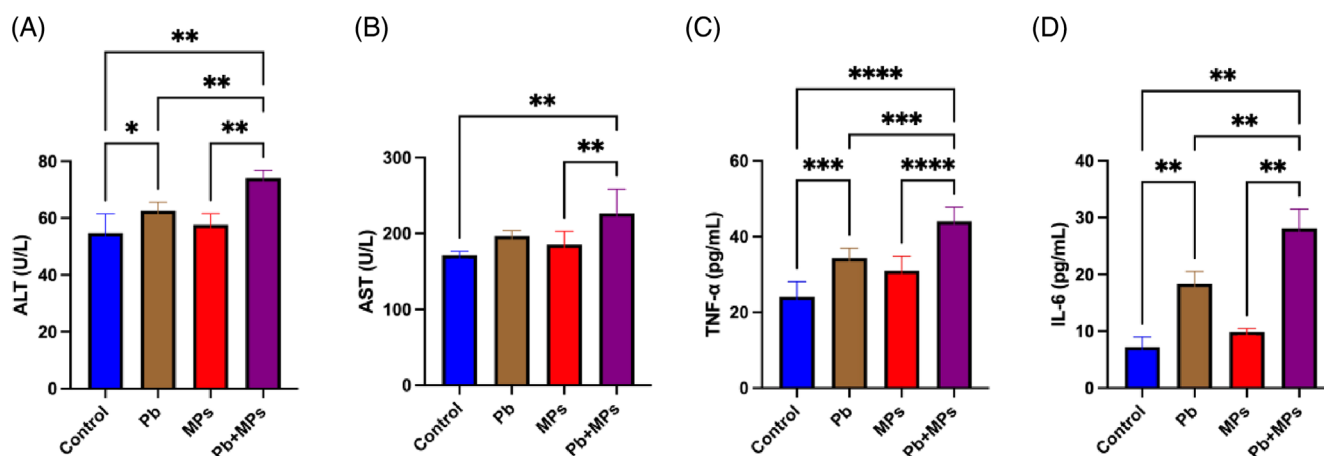


FIGURE 5 Effects of lead (Pb), microplastics (MPs), and their co-exposure on serum biochemical indicators of liver function and inflammatory markers in mice after 6 weeks. (A) The levels of serum alanine aminotransferase (ALT), (B) the levels of serum aspartate amino transferase (AST), (C) the levels of liver tumor necrosis factor-alpha (TNF- α), and (D) liver interleukin-6 (IL-6) levels. Data are expressed as the mean $\pm$ SD, $n = 6$ in each group. * $p < .05$; ** $p < .01$ significant different compared with the control group.

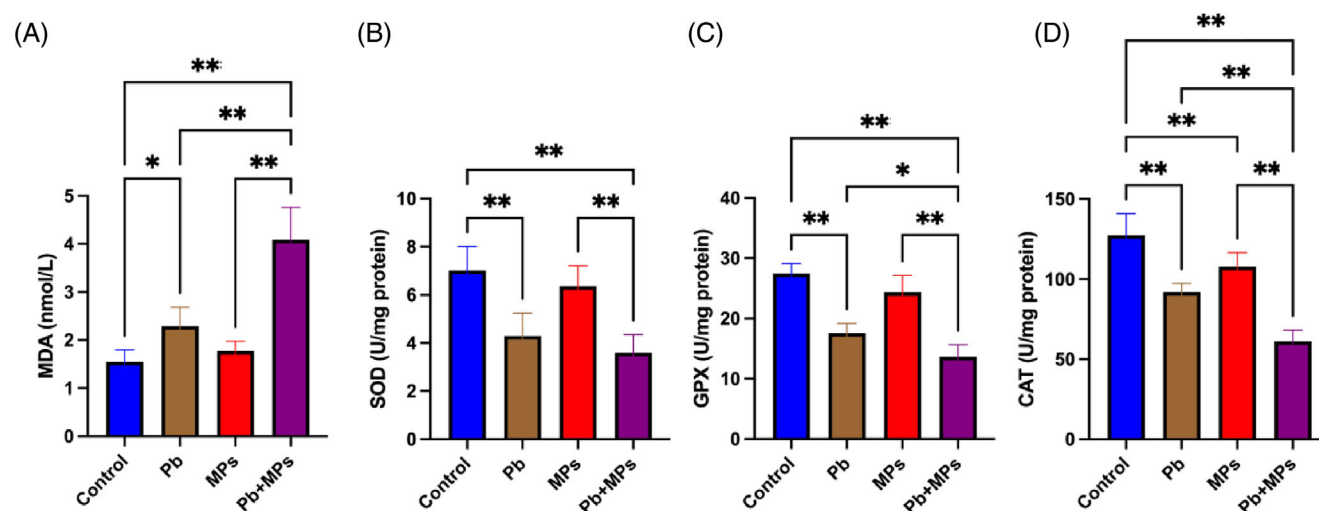


FIGURE 6 Effects of lead (Pb), microplastics (MPs), and their co-exposure on oxidative stress/antioxidant markers in the liver of mice. (A) Malondialdehyde (MDA), (B) superoxide dismutase (SOD), (C) glutathione peroxidase (GPx), and (D) catalase (CAT). Data are expressed as the mean $\pm$ SD ($n = 6$). * $p < .05$ and ** $p < .01$ significant different compared with the control group.

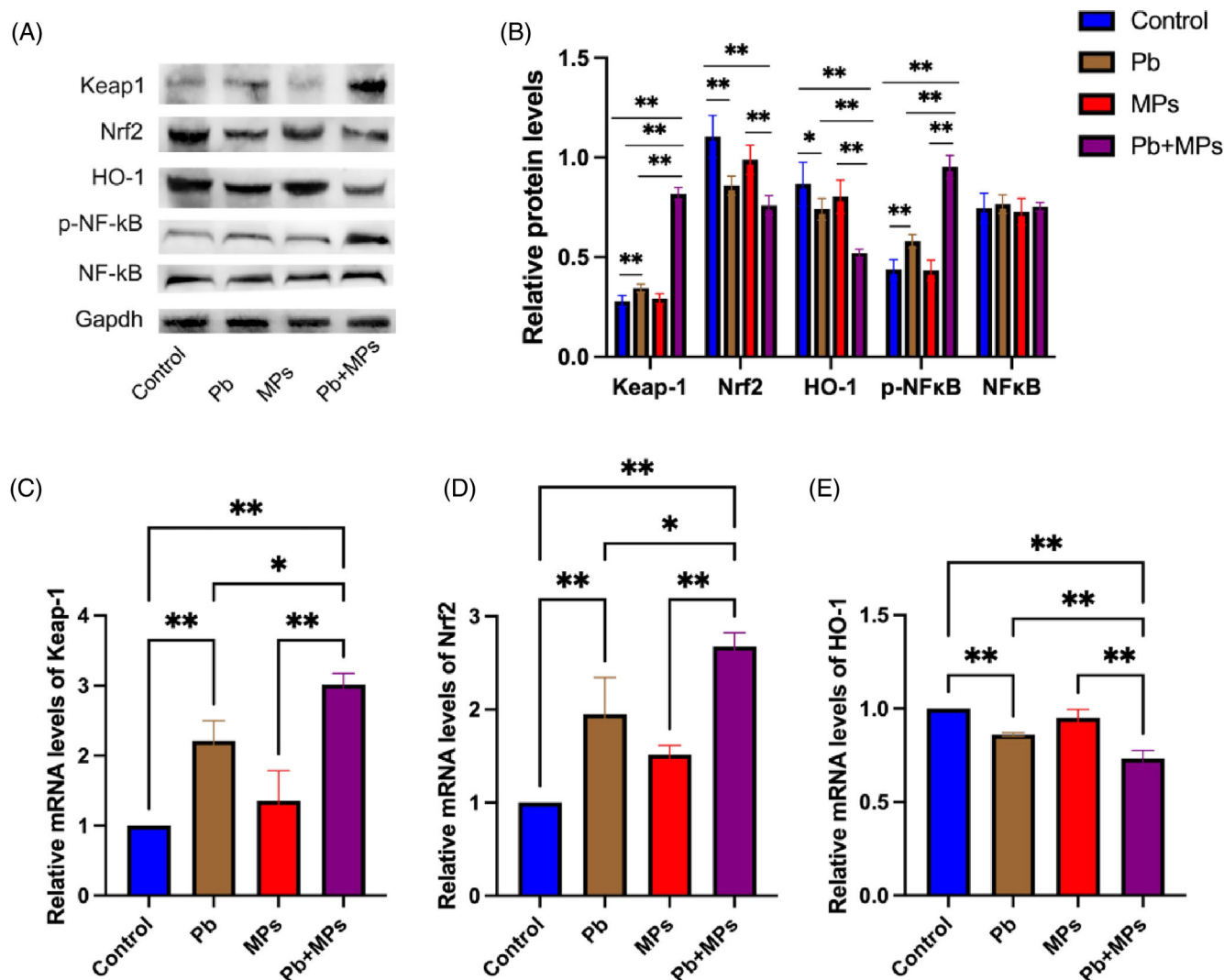


FIGURE 7 Effect of lead (Pb) and microplastics (MPs) in the hepatic Keap1/Nrf2 and NF-κB signal pathway. (A) Protein expression of Keap1, Nrf2, HO-1, p-NF-κB, and NF-κB induced by administration of Pb and MPs in liver. (B) Protein expression was normalized against GAPDH. (C) The Keap1, Nrf2, and HO-1 mRNA expression ($n = 3$). The respective bars illustrate the mean $\pm$ S.D. * $p < .05$ and ** $p < .01$ significant different compared with the control group.

and HO-1. Compared with control mice, the mRNA levels of Keap1 and Nrf2 were increased Pb and Pb + MPs mice, but they were decreased in Pb and MPs mice than Pb + MPs mice. In contrast, HO-1 was observed to inhibited by Pb and Pb + MPs treatment than the control treatment, but promoted in Pb and MPs mice than Pb + MPs mice.

4 | DISCUSSION

The diabetes prevalence continues to increase rapidly worldwide, especially in low- and middle-income countries.^{24,25} Although a continuing increase awareness of diabetes was found in self-reported, the diabetes percentage control in 2017 remained the same stage as in 2013;²⁶ which demonstrates that the prevention of diabetes needs further improvement and strengthening. In addition to genetic and

lifestyle factors, the causes of diabetes are also associated with environmental factors, such as excessive exposure to certain toxins and chemicals may contribute the development of diabetes.²⁷ Although numerous researches have investigated the relationships between T2DM and exposure to individual metals, the effect of metal and MPs exposure and the underlying molecular mechanisms that contribute to T2DM pathogenesis remain scarce in this scope.

In the present study, we provide the combined or single exposed to Pb and two types of MPs (PVC and PE) effects on glucose metabolism and explore the role of Nrf2/NF-κB pathway in mediating these effects. The findings of the present study indicate that under the current exposure duration (6 weeks) and dosage levels, exposure to Pb and MPs alone or combination have no significant impact on the weekly body weight, weekly fasting blood glucose, and insulin secretion. However, treatment with Pb + MPs may potentially affect GSP, which reflect the average blood glucose levels over a certain period

and are not influenced by fluctuations of blood glucose levels. This indicates that combined exposure to Pb and MPs may influence glucose metabolism.

The primary criteria for assessing glucose homeostasis includes glucose tolerance, fasting blood glucose, and insulin. To further elucidate the specific roles of Pb and MPs on glucose metabolism, we also conducted ipGTT and ipITT. Studies have shown that increases in glucose AUC of ipGTT and ipITT, and increases in HOMA-IR were observed both after Pb + MPs exposure, suggesting that combined exposure of Pb and MPs synergistically promotes insulin resistance. HOMA-IR index is used for evaluation of insulin resistance, which has a great relationship with the results of glucose clamp measurement in subjects without significant hyperglycemia.²⁸ HOMA- β index is used to quantifies β -cell function and mass,²⁹ which was observed that there had no significant difference changes in the four groups of the present study.

This indicates that the combined exposure to Pb and MPs does not have a significant effect on pancreatic secretory function, and that also suggests that their target of action maybe in the liver. The present study also confirmed the effects of Pb and MPs on the pancreatic β cell and hepatic damage. Our data suggest that although co-exposure to Pb and MPs affects the acinar cells of the pancreas, it does not impact the islets of Langerhans. However, for the liver, both individual exposure to Pb and MPs can cause varying degrees damage in the structure of the hepatocytes. It is noteworthy that when combined exposure to both, the more severe liver injury was observed.

Several reports have documented the relationship between Pb exposure and glucose dys-metabolism, as well as complications related to diabetes.^{7,30,31} However, we observed that exposures to Pb and MPs alone had no statistically significant difference on insulin resistance, although the Pb exposure showed a tendency to increase insulin resistance. This may be attributed to the short duration of exposure in our experiments, as well as the exposure dosage and the strain of mice used. Previous finding suggested that fed with PS with high-fat diet may cause metabolic disturbances, including increased fat accumulation in the liver and deterioration of glucose tolerance.¹⁰ Further research has indicated that exposure to PS (with a diameter of 1 μ m) for 2 weeks may eventually leading to insulin resistance and even diabetes.³²

In general, apart from polymer type, exposure duration, and concentration, the toxicity of MPs also largely depends on particle size. Considering smaller particles (<1.5 μ m) have a higher likelihood of being absorbed and accumulating in the tissues,³³ studies generally focus on nano-sized plastics. However, recent studies reported that MPs with larger than 5 μ m actually constituted a considerable proportion detected in human samples.^{34,35} PVC and PE both as the most common MPs, which widely used in many sectors and consumer products, however, few studies have focused the toxicity of them in vivo. In the current study, we selected the mixture of PE and PVC (both with a diameter of 6.5 μ m), with the dosage according to the daily real-world human exposure.³²

The present study also investigated lipids serum levels, and found that apart from an increase of LDL-c in Pb-MPs mice, neither Pb nor MPs had a significant impact on TG, CHOL, and HDL-c levels. However, regarding liver function (AST and ALT), inflammation markers

(TNF- α and IL-6), and oxidative stress-related indicators (MDA, SOD, GPx, and CAT); exposure to Pb alone or combined exposure to Pb and MPs exhibited significant adverse effects. This suggests that oxidative stress responses and inflammation are related to the hepatic glucose metabolism disturbances caused by Pb and MPs. In addition, to validate the above-mentioned studies, hepatic Keap1/Nrf2 and NF- κ B signal relative protein were conducted. The results indicate that the Keap1 and p-NF κ B protein levels were increased but the Nrf2 and HO-1 protein levels were decreased in Pb and Pb + MPs groups when compared with the control group, and Keap1 mRNA and Nrf2 mRNA were upregulated, but HO-1 mRNA was downregulated.

Nrf2 is a key transcription factor that enhances cellular antioxidant capacity by promoting the synthesis of GSH, CAT, GPx, SOD, and HO-1. Additionally, Nrf2 acts as a potent antiinflammatory factor by suppress pro-inflammatory cytokine production through the inhibition of NF- κ B activation.³⁶ It is worth noting that the interaction between Nrf2 and cytoplasmic Keap1 can regulate the Nrf2 transcriptional activity. Keap1 plays a role in facilitating the proteasome-mediated Nrf2 degradation, primarily within the cytoplasm.^{37,38} Based on our findings, co-exposure to Pb and MPs may initially lead to inflammation and oxidative stress. This activation subsequently increases the expression of Keap1, which, in turn, enhances the transcription of Nrf2 mRNA. However, prolonged stimulation also results in increased depletion of Nrf2 protein. This imbalance further stimulates the increased expression of NF- κ B, ultimately leading to disturbances in liver glucose metabolism.

5 | CONCLUSIONS

Associated with all changes mentioned above, exposure to Pb can increase liver inflammation and oxidative stress levels, but under the current dosage and duration conditions, it may not reach the extent of affecting glucose metabolism. However, when co-exposed with MPs, it can amplify the extent of liver damage and impact glucose metabolism through the NF- κ B/Keap1/Nrf2 signaling pathway.

ACKNOWLEDGMENTS

The authors would like to thank Dr. Xinxin Yang from Xuzhou Medical University for kindly provide us animals and necessary instrument.

FUNDING INFORMATION

This work was supported by Shandong Provincial Medicine and Health Technology Development Program [202103061021]; the Key Research and Development Program of Linyi City (Medical) Project [2022YX0112] and [2022YX0109].

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Haibo Yang  <https://orcid.org/0000-0002-5715-0849>

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How to cite this article: Zhu M, Li P, Xu T, et al. Combined exposure to lead and microplastics increased risk of glucose metabolism in mice via the Nrf2/NF-κB pathway. *Environmental Toxicology*. 2024;39(5):2502–2511. doi:10.1002/tox.24125